

Week 7 Tutorial

Estimating a Firm's Cost of Capital

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Agenda

- 4-7
- 4-9
- 4-12
- 4-14
- 4-16
- 4-17

Knowledge review 1: Key formulas

Bond Price

$$= \frac{[\text{Interest} + \text{Principal}] \times (1 - Pb) + [\text{Interest} + \text{Principal}] \times Pb \times Re}{(1 + k_d)}$$

Levered beta

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_e w_e$$

$$\text{Firm Value}_0 = \sum_{t=1}^N \frac{E(\text{Firm FCF}_t)}{(1 + WACC)^t}$$

Knowledge review 1: Key formulas

When using the CAPM to estimate the cost of equity, it is standard practice to estimate the firm's equity beta by computing averages of equity betas of comparable firms with similar operating risk characteristics. Because equity betas vary with the use of financial leverage, we need to adjust the betas of the comparison firms for differences in how the firms are financed. This requires an understanding of the relationship between financial leverage and beta.

To develop an understanding of this relationship, we begin by noting that the beta for a firm is simply a weighted average of the betas of each of the sources of capital in the firm's capital structure. So the beta for firm j can be defined as follows:

Unlevered beta

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right) \quad (1)$$

Because firm value = equity value + debt value, we can solve for the equity beta by solving equation (1) for β_{Equity_j} as follows:

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) \quad (2)$$

Knowledge review 1: Key formulas

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right) \quad (1)$$

- $FV = D + E$
- Step 1 (multiply both sides by $D + E$): $\beta_{\text{Firm}}(D + E) = \beta_{\text{Equity}}E + \beta_{\text{Debt}}D$
- Step 2 (isolate the $\beta_{\text{Equity}}E$ term): $\beta_{\text{Equity}}E = \beta_{\text{Firm}}(D + E) - \beta_{\text{Debt}}D$
- Step 3 (divide both sides by E): $\beta_{\text{Equity}} = \beta_{\text{Firm}} \frac{D+E}{E} - \beta_{\text{Debt}} \frac{D}{E}$, $\beta_{\text{Equity}} = \beta_{\text{Firm}} \left(1 + \frac{D}{E} \right) - \beta_{\text{Debt}} \frac{D}{E}$
- Here, $\frac{D}{E}$ represents the Debt/Equity (Capitalization) ratio

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) \quad (2)$$



Problem 1

4-7 ESTIMATING THE UNLEVERED BETA The CFO of Sterling Chemical is interested in evaluating the cost of equity capital for his firm. However, Sterling uses very little debt in its capital structure (the firm's debt-to-equity capitalization ratio is only 20%), while larger chemical firms use substantially higher amounts of debt. The following table shows the levered equity betas, debt-to-equity ratios, and debt betas for three of the largest chemical firms:

Company Name	Levered Equity Betas	Debt/Equity Capitalization	$\frac{D}{E}$
			Assumed Debt Betas
Eastman Chemical Co. (EMN)	1.79	30.77%	0.30
Celanese Corp. (CE)	1.98	23.55%	0.30
Dow Chemical Company (DOW)	1.71	21.60%	0.30

- Use the information given above to estimate the unlevered equity betas for each of the companies.
- If Sterling's debt-to-equity capitalization ratio is .20 and its debt beta is .30, what is your estimate of the firm's levered equity beta?

Problem 1

Answer

$$\beta_{Firm_j} = \beta_{Equity_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{Debt_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right)$$

- Let: $\frac{D}{E} = x$, Then: $D = Ex$
 - So: Equity weight = $\frac{E}{D+E} = \frac{1}{x+1}$, Debt weight = $\frac{D}{D+E} = 1 - \frac{1}{x+1} = \frac{x}{x+1}$
 - $\beta_{Firm} = \beta_{Equity} \frac{1}{x+1} + \beta_{Debt} \frac{x}{x+1} = \frac{\beta_{Equity} + x\beta_{Debt}}{x+1}$
1. $\beta_{Firm-EMN} = \frac{1.79+0.3 \times 30.77\%}{1+30.77\%} = 1.4394$; $\beta_{Firm-CE} = \frac{1.98+0.3 \times 23.55\%}{1+23.55\%} = 1.6598$; $\beta_{Firm-DOW} = \frac{1.98+0.3 \times 21.60\%}{1+21.60\%} = 1.4595$; $\beta_{Firm} = \frac{1.4394+1.6598+1.4595}{3} = 1.5196$

Problem 1

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)$$

2. $\beta_{\text{Equity-Sterling}} = 1.5196 \times (1 + 0.2) - 0.3 \times 0.2 = 1.7635$

Knowledge review 2: YTM

- Promised yield to maturity (YTM)

$$\text{Bond Price} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})}$$

- Expected YTM (k_d)

$$\text{Bond Price} = \frac{[\text{Interest} + \text{Principal}] \times (1 - Pb) + [\text{Interest} + \text{Principal}] \times Pb \times Re}{(1 + k_d)}$$

- when $Pb = 0$, $\text{YTM} = k_d$, Discount rate = WACC
- when $Pb > 0$, $\text{YTM} > k_d$, Discount rate $>$ WACC

Note: Unless the problem explicitly mentions default risk, you should assume there is no possibility of default.

Table 4A-1 Average Cumulative Default Rates for 1 to 10 Years: 1983–2013

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Investment grade	0.10%	0.30%	0.56%	0.86%	1.19%	1.52%	1.86%	2.21%	2.54%	2.89%
Speculative grade	4.67%	9.66%	14.66%	18.69%	22.35%	25.58%	28.43%	30.91%	33.13%	35.13%

Source: Moody's Investors Service, *Corporate Default Study: Corporate Default and Recovery Rates, 1920–2013*, Exhibit 34, p. 34.

Investment grade

Speculative grade
"Junk bonds"

	BBB+	BBB	BBB-	Below investment grade
Baa2	BBB	BBB		
Baa3	BBB-	BBB-		
Ba1	BB+	BB+		Noninvestment grade
Ba2	BB	BB		Speculative
Ba3	BB-	BB-		
B1	B+	B+		Highly speculative
B2	B	B		
B3	B-	B-		
Caa1	CCC+	CCC		Substantial risk
Caa2	CCC	—		In poor standing
Caa3	CCC-	—		
Ca	—	—		Extremely speculative
C	—	—		May be in default
—	—	DDD		Default
—	—	DD		
—	D	D		

$$(1000 \times 14\% + 1000) \times 50\%$$

11 scenarios: 10 default, 1 non-default

4: *sum*(3)

11.87%

Expected YTM = Cost of debt

Promised YTM

Problem 2

4-9 CALCULATING THE EXPECTED YTM International Tile Importers Inc. is a rapidly growing firm that imports and markets floor tiles from around the world. The tiles are used in the construction of custom homes and commercial buildings. The firm has grown so fast that its management is considering the issuance of a five-year interest-only note. The notes would have a principal amount of \$1,000 and pay 12% interest annually, with the principal amount due at the end of year 5. The firm's investment banker has agreed to help the firm place the notes and has estimated that they can be sold for \$800 each under today's market conditions.

- a. What is the promised YTM based on the terms suggested by the investment banker?
- b. Refer to the appendix to this chapter for this analysis. The firm's management looked with dismay at the YTM estimated in Problem 4-9(a) because it was much higher than the 12% coupon rate, which is much higher than the yield to maturity currently paid on investment-grade debt. The investment banker explained that, for a small firm such as International Tile, the bond rating would probably be in the middle of the speculative grades, which requires a much higher yield to attract investors. The banker even suggested that the firm recalculate the expected YTM on the debt under the following assumptions: The risk of default in years 1 through 5 is 5% per year, and the recovery rate in the event of default is only 50%. What is the expected YTM under these conditions?

Problem 2

Answer

$$\text{Bond Price} = \frac{\text{Interest} + \text{Principal}}{(1 + YTM)}$$

$$a. \quad 800 = \frac{1000 \times 0.12}{1 + YTM} + \frac{1000 \times 0.12}{(1 + YTM)^2} + \dots + \frac{1000 \times 0.12 + 1000}{(1 + YTM)^5}, YTM = 18.46\%$$

Bond Price

$$= \frac{[\text{Interest} + \text{Principal}] \times (1 - Pb) + [\text{Interest} + \text{Principal}] \times Pb \times Re}{(1 + k_d)}$$

b. [Excel demonstration in class](#)

Hints: use Excel's Rate and IRR functions.

Problem 3

4-12 THREE-STEP PROCESS FOR ESTIMATING A FIRM'S WACC Harriston Electronics builds circuit boards for a variety of applications in industrial equipment. The firm was founded in 1986 by two electrical engineers who left their jobs with General Electric (GE) Corporation. Harriston Electronics' balance sheet for year-end 2014 describes a firm with \$1,184,841,000 in assets (book values) and invested capital of approximately \$2.2 billion (based on market values):

	December 31, 2014	
	Balance Sheet (Book Values)	Invested Capital (Market Values)
Liabilities and Owners' Capital		
Current liabilities		
Accounts payable	\$ 17,550,000	
Notes payable	20,000,000	20,000,000
Other current liabilities	22,266,000	
Total current liabilities	\$ 59,816,000	\$ 20,000,000
Long-term debt (7.5% interest paid semiannually, due in 2020)	\$ 650,000,000	\$ 624,385,826
Total liabilities	\$ 709,816,000	\$ 644,385,826
Owners' capital		
Common stock (\$1 par value per share)	\$ 20,000,000	
Paid-in-capital	200,025,000	
Accumulated earnings	255,000,000	
Total owners' capital	\$ 475,025,000	\$1,560,000,000
Total liabilities and owners' capital	\$1,184,841,000	\$2,204,385,826

Harriston's CFO, Margaret L. Hines, is concerned that its new investments be required to meet an appropriate cost of capital hurdle before capital is committed. Consequently, she initiated a cost of capital study by one of her senior financial analysts, Jack Frist. Shortly after receiving the assignment, Frist called the firm's investment banker to get input on current capital costs.

Frist learned that, although the firm's current debt capital required a 7.5% coupon rate of interest (with annual interest payments and no principal repayments until 2020), the current yield to maturity on similar debt had risen to 8.5% so that the current

market value of the firm's outstanding bonds had fallen to \$624,385,826. Because the firm's short-term notes were issued within the last thirty days, the 9% contract rate on the notes was the same as the current cost of credit for such notes.

- What are Harriston's total invested capital and capital structure weights for debt and equity? (*Hint:* The firm has some short-term debt [notes payable] that, like long-term debt, is also interest-bearing debt.)
- Assuming a long-term US Treasury bond yield of 5.42% and an estimated market risk premium of 5%, what is Harriston's cost of equity based on CAPM if the firm's levered equity beta is 1.2?
- What is your estimate of Harriston's WACC? The firm's tax rate is 35%.

Problem 3

Answer

$$a. \quad \text{Total invested capital} = 2,204,385,826; \quad \frac{\text{Short-term debt}}{\text{Enterprise value}} = \frac{20,000,000}{2,204,385,826} \times 100\% =$$

$$0.91\%, \quad \frac{\text{long-term debt}}{\text{Enterprise value}} = \frac{624,385,826}{2,204,385,826} \times 100\% = 28.32\%, \quad \frac{\text{Equity}}{\text{Enterprise value}} =$$

$$\frac{1,560,000,000}{2,204,385,826} = 71\%$$

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$b. \quad k_e = 5.42\% + 1.2 \times 5\% = 11.42\%$$

Problem 3

$$WACC = k_d(1 - T)w_d + k_pw_p + k_e w_e$$

c. $WACC = 9\% \times (1 - 35\%) \times 0.91\% + 8.5\% \times (1 - 35\%) \times 28.32\% + 11.42\% \times 71\% = 9.7\%$

Knowledge review 3: Fama-French three-factor model

- CAPM often **under-predicts** returns for small-cap and value stocks. Fama and French (1993) added these two factors after finding consistent anomalies in real market returns that CAPM couldn't explain.

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$k_e = k_{rf} + b \times (\text{ERP}) + s \times (\text{SMBP}) + h \times (\text{HMLP})$$

Factor Models A second approach that was introduced in the 1980s is the use of multifactor risk models that capture the risk of investments with multiple betas and factor risk premiums. These risk factors can come from macroeconomic variables, like changes in interest rates, inflation, or gross domestic product (GDP), or from the returns of what are known as factor portfolios. Factor portfolios can be formed using purely statistical procedures, like factor analysis or principal components analysis, or by grouping stocks based on their characteristics.

For example, the *Fama-French three-factor model* captures the determinants of equity returns using three risk premiums:²² the market risk premium of CAPM, a size risk premium, and a risk premium related to the relative value of the firm when compared to its book value (historical cost-based value). The Fama-French equation for the cost of equity capital includes three factors with their associated risk premiums (thus the use of the term *three-factor model*), as shown in Equation (4.9):

$$k_e = k_{rf} + b \times (\text{ERP}) + s \times (\text{SMBP}) + h \times (\text{HMLP}) \quad (4.9)$$

where

k_e = the required rate of return for the common stock of the firm

k_{rf} = the risk free rate of interest (assumed to be 3%)

b , s , and h are estimated coefficients for the particular firm whose cost of equity is being evaluated

ERP = equity risk premium, equal to the difference in the expected rate of return from investing in the market portfolio of equities and the long-term risk-free rate of interest (as noted earlier, we use an ERP of 5%)

SMBP = the small-minus big-risk premium, estimated from historical return differences in large- and small-cap stocks (3.23%)

HMLP = the high-minus low-risk premium, estimated as the difference between the historical average annual returns on the high book-to-market (value) and low book-to-market (growth) portfolios (4.08%).²³

Knowledge review 3: Fama-French three-factor model

Table 4-2 Fama-French Versus CAPM Estimates of the Cost of Equity

Company Name	Fama-French Coefficients			FF Cost of Equity	CAPM Beta	CAPM Cost of Equity	CAPM Minus
	<i>b</i>	<i>s</i>	<i>h</i>				FF Cost of Equity
Apple Inc	0.9716	0.1586	(0.5859)	5.98%	0.82	7.10%	1.12%
IBM	0.6801	(0.0017)	(0.5720)	3.84%	0.49	5.43%	1.59%
Merck & Co	0.7705	(1.0911)	(0.0308)	3.20%	0.52	5.58%	2.38%
Pepsico	0.4957	(0.4501)	(0.0300)	3.90%	0.37	4.83%	0.92%
Pfizer	0.9404	(1.1152)	(0.3177)	5.40%	0.78	6.88%	1.48%
Wal Mart	0.1597	0.2979	(0.2502)	5.78%	0.35	4.74%	−1.04%

Problem 4

4-14 ESTIMATING THE COST OF EQUITY USING THE FAMA-FRENCH METHOD Telecom services is an industry under rapid transformation because telephone, Internet, and television services are being brought together under a common technology. In fall 2006, the telecom analyst for HML Capital, a private investment company, was trying to evaluate the cost of equity for two giants in the telecom industry: SBC Communications (AT&T) and Verizon Communications. Specifically, he wanted to look at two alternative methods for making the estimate: the CAPM and Fama-French three-factor model. The CAPM utilizes only one risk premium for the market as a whole, whereas the Fama-French model uses three (one for each of three factors). The factors are (1) a market risk premium, (2) a risk premium related to firm size, and (3) a market-to-book risk premium. Data for the risk premium sensitivities (b , s , and h) as well as the beta coefficient for the CAPM are listed in the following table:

Company Name	Fama-French (FF) Coefficients			CAPM Beta
	b	s	h	
SBC Communications (AT&T)	1.0603	-1.4998	1.0776	0.62
Verizon Communications	1.1113	-0.9541	0.639	0.79

The risk premia for each of the risk factors are:

Coefficients	Risk Premia
Market risk premium	5.00%
Size risk premium	3.36%
Market-to-book risk premium	4.40%

- If the risk-free rate of interest is 3%, what is the estimated cost of equity for the two firms using the CAPM?
- What is the estimated cost of equity capital for the two firms using the Fama-French three-factor model? Interpret the meaning of the signs ($\pm$) attached to each of the factors (i.e., b , s , and h).

Problem 4

Answer

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$a. \quad k_{e-ATT} = 3\% + 0.62 \times 5\% = 6.1\%, k_{e-verizon} = 3\% + 0.79 \times 5\% = 6.95\%$$

$$k_e = k_{rf} + b \times (\text{ERP}) + s \times (\text{SMBP}) + h \times (\text{HMLP})$$

$$b. \quad k_{e-ATT} = 3\% + 1.0603 \times 5\% - 1.4998 \times 3.36\% + 1.0776 \times 4.4\% = 8\%, k_{e-ATT} = 3\% + 1.1113 \times 5\% - 0.9541 \times 3.36\% + 0.639 \times 4.4\% = 8.16\%$$

The coefficients on a particular Fama French factor indicate the level of exposure of the company to that factor. Low (high) values of the coefficient imply low (high) exposures. A negative sign implies that including the company in a portfolio reduces the risk to the corresponding Fama French factor.

Problem 5

4-16 PROMISED VERSUS EXPECTED YTM This problem is based on the appendix to this chapter. In May 2005, the credit rating agencies downgraded the debt of General Motors Corporation (GM) to junk status.

- a. Discuss the effect of this drop in credit status on the company's WACC.
- b. If you have been using the YTM for the firm's bonds as your estimate of the cost of debt capital and you use the same estimation procedure following the downgrade, has your estimate become more or less reliable? Explain.

Problem 5

Answer

- a. The downgrade to junk status reduced the market price of GM's bonds, thereby **increasing** their yield to maturity.
- b. However, as we learned in the appendix to Chapter 4, the calculated YTM reflects the **promised** yield, not the investor's expected yield for below-investment grade bonds. Consequently, the YTM **overstates** the investor's expected rate of return, and its use in the calculation of GM's WACC will result in an **upwardly** biased estimate.

Problem 6

4-17 COMPREHENSIVE WACC COMPUTATION On behalf of your firm, you are evaluating a privately held takeover target. The table below contains cost of capital metrics for two comparable publicly traded firms that are in the same general business and are similar in size compared to the acquisition target. You decide to equally weight all the comp data to estimate the correct discount rate to use in valuing the target. The target will be financed with 20% debt, with a pretax cost of debt equal to 4%, and the debt beta is assumed to equal 0. Assume that all of the firms in your analysis are in the 35% tax bracket and that all of the firms have a policy to leave their debt ratio constant. Answer the following questions regarding the correct cost of capital to use in valuing the target. For all parts of this problem, assume a risk-free rate of 3% and a market risk premium of 6%.

Comp Company	Equity Beta	Debt Beta	D/E
Comp 1	1.5	2	15%
Comp 2	2.1	2	30%

- What is the estimated firm or asset (unlevered equity) beta for the target firm based on the comp data?
- What is the estimated levered cost of equity for the target based on the comp data and the target's debt/value ratio of 20%?
- Using the equity cost already calculated and the information on the cost of debt in the problem, what is the WACC to use in discounting the target's projected firm free cash flows?

Problem 6

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j} \left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j} \right) + \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j} \right)$$

$$\text{Let: } \frac{D}{E} = x, \quad \beta_{\text{Firm}} = \frac{\beta_{\text{Equity}} + x\beta_{\text{Debt}}}{x + 1}$$

$$a. \quad \beta_{\text{Firm-Comp1}} = \frac{1.5 + 15\% \times 2}{15\% + 1} = 1.3304, \quad \beta_{\text{Firm-Comp2}} = \frac{2.1 + 30\% \times 2}{30\% + 1} =$$

$$1.6615, \quad \overline{\beta_{\text{Firm}}} = \frac{1.3304 + 1.6615}{2} = 1.4960$$

Problem 6

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)$$

b. $\frac{D}{E} = \frac{D}{FV} \div \frac{E}{FV} = \frac{20\%}{80\%} = 25\%$, $\beta_{\text{Equity-targeted}} = 1.4960 \times (1 + 25\%) - 0 \times 25\% = 1.87$

$$k_e = k_{rf} + \beta_e(k_m - k_{rf})$$

$$WACC = k_d(1 - T)w_d + k_pw_p + k_ew_e$$

c. $WACC = 4\% \times (1 - 35\%) \times 20\% + (3\% + 1.87 \times 6\%) \times 80\% = 11.90\%$